

# ActInf Livestream #004.1 "Cultural Affordances: Scaffolding Local Worlds"

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live streaming hello everyone and welcome this is team com podcast 4.1 on september 15th 2020. welcome to everybody hello all participants i'm going to share my screen and we will jump right in this is team com podcast 4.1 and this is team com teamcom is an experiment in online team communication and learning related to active inference you can find us on twitter at inferenceactive you can email us you can find us on the public key base team as well as at the youtube channel this is a recorded and archived live stream so please provide us with any feedback participants as well as viewers so that we can improve our work all backgrounds and perspectives are welcome here and also just on a note of video etiquette if people are muted in a noisy background or raising their hands when there's this many participants in the chat it would be excellent and help us with our turn taking and reduction of uncertainty about who's to speak next so here we are in stream 4.1 and the structure of today is going to go like this first we're going to have a few check-in questions and warm-up discussion then we'll get to the paper itself which is ramstead at all 2016. we'll talk about the big aims of the paper go through a roadmap of the paper to see how the authors get from a to z we'll go through the abstract to understand how the authors represent their own work then we'll go through the box of definitions as well as the figures at that point we'll have time for questions and other insights and also we've implicitly prepared for a podcast 4.2 where we'll be able to have further discussions on some key quotations or anything arising that we don't get to this week so thanks for that part here we are in the check-in and everybody can feel free to introduce themselves in their location if they would like the first question is pretty broad but that's what the warm-up is about so what is culture it's a good question or in the sense of this paper in the sense of culture that's being scaffolded within a kind of framework that we might want to build out what would that kind of culture refer to well so i'm maxwell ramstad i'm uh the first author on this paper cultural affordances um by the way i'm really grateful uh for this uh podcast you all have been so great just in general over the last few weeks and i'm i'm particularly touched

that you're giving uh the work that we've developed so much attention um and it's like just i'm just so full of gratitude this is so cool um so culture at least in the sense that we're using uh the term here means something like the taken for granted i think is sort of the heuristic way that i like to think about it it's sort of like in a given local frame of reference uh in a given environment shared by a group of people a living together what are the things that are taken for granted what are the sets of expectations and beliefs and sets of practices and norms um you know that that are shared and that essentially uh structure people's intuitive grip on their shared social world any other thoughts on culture there yeah in sociology uh we define culture it's kind of three different dimensions you have the symbolic ideal dimension which is really i think equivalent to or equivalent to semantic knowledge uh we break it up in terms of norms values and beliefs uh we have a psychobehavioral dimension which is really about in body culture how your body moves um the skills that you have in in terms of psychological kind of terminology i would really say your skills and habits um in in kind of that non-declarative memory sense and then the third dimension is really our material dimensions of the physical things that we create the tools the buildings the fashion um which i think the degree to which i understand the affordance literature i think all three of those dimensions definitely have versions of affordances that could connect to that um i think that was one thing i did see in the paper that it did seem to be very much focused on just that semantic kind of symbolic dimension and not really uh explicitly incorporating the other dimensions that at least in sociology we would kind of um include in that definition of culture cool yeah thanks for that that was very rich actually uh very interesting uh points that you're raising yeah i should just give some maybe some historical background on this paper this was the first in a series of papers that have been developed into basically like a a variational or free energy approach to culture and i think a lot of the elements that you're mentioning were only sketched in this paper uh and were developed in i think further iterations of the model uh so everything that you're that you just outlined uh richard i think are are very essential and crucial things that need to be tackled when studying culture so it's great that you're raising them cool and i think the question about multiple kinds of culture ranging from the semantic concepts to the externalized productions or the artifacts of culture bring us this idea of scaffolding culture so what is a way that culture is scaffolded someone can raise their hand maybe someone new but something that guides the construction just like a scaffold i'll give one way which is education because pedagogy is sort of the guide rails for how individuals learn about those norms as well as how to use the artifacts and construct new artifacts these are ways in which culture is

scaffolded because it provides structure for the growth to happen within any other comments on that okay and then the last warm-up question is what is shared intentionality or what's a way in which somebody has seen shared intentionality play out in their recent experience well so there are two parts to that right there's shared and there's intentionality so intentionality at least in the sense that we're using it in the paper draws on the philosophical construct of intentionality that we we can basically uh you know you can draw a lineage all the way back to brentano effectively but so by intentionality we mean orientation towards the world uh we don't necessarily mean intentionality in the sense of uh having the intention to do this or that it really it's from the latin i think intensio which is the word that was used to describe what a uh what an archer is doing with his arrow uh to hit the target it is intensio with the target and so it it it's this that we're trying to get at and one of the things that we're trying to argue here is that what part of what makes um a shared social world just that a shared social world is that there are forms of relating to that world so forms of intentionality that are also shared uh among the the people who live in that world together so that's that's what we were trying to get at ultimately i think and i think what's interesting in terms of if you take that definition and you look back at the scaffolding um you know if you look within the individual there's this network of semantic and episodic memory and non-declarative memory that's organized in the brain if you put me in contact with another individual from my culture who shares my language and we begin to construct this we space if you look at the phenomenology um and then now what happens with this dyad either we introduce a third person or we then one of us leaves the relationship and we go uh interact with someone else if you spread that dyadic interaction out to a network and then you look at the fractal scaling of that network there's this way to i think to very deliberately keep track and build up from organized memory and brain to an institution of 25 000 people dynamically interacting by following this this these mechanisms by which you create the shared space and looking at those emerging properties and trying to model them mathematically couldn't agree and there's a quote from the paper which i'll spare maxwell from reading is uh near the end it says to address more complex social situations it is useful to revise current sociocognitive models of joint attention to encompass fundamentally triadic situations in which the third is the socially constituted niche of affordances supported by local ontologies and abilities so we're just jumping in today but that's sort of where we're wanting to go with this paper and in discussion is towards understanding what is that third space and how does the aim or does the intent play in yeah stephen yeah and i think this cultural thing also brings in that into subjectivity it gives a way to

understand how to approach it beyond education beyond sort of the banking of programming people saying well what's in the environment particularly now that we've got situations of cross-cultural what cultures have to try and understand and work together is it all about teaching us how to tolerate each other or is it about understanding how we perform in the environment so i think it's got a lot of well here we are finally at the paper uh cultural affordances scaffolding local worlds through shared intentionality and regimes of attention so we'll definitely want to unpack each of these concepts and it's in frontiers in psychology from 2016. the big aim of the paper grasped from the abstract in the introduction section is to better understand how culture and context interact with human biology to shape human behavior cognition and experience so kind of all of it kind of things that richard was bringing up early on and from the introduction the integrative framework we propose bridges cognitive and social sciences to provide two things first an expanded concept of affordance that extends to socio-cultural forms of life which will unpack in box one and two a multi-level account of the socio-culturally scaffolded forms of affordance learning and the transmission of affordances in pattern sociocultural practices so the key piece there is that multi-level account because we're not going to have an individual level account by which collective outcomes arise or a group level account by which individuals are shaped in their behavior uniquely those are sort of two sides of the same coin or some of the sides of a many-sided coin that constitute this top down and bottom-up relationship that we see in a lot of multi-scale systems like the brain being nested within a body inside of a niche inside of these institutions that last beyond the lifespan of the organism any thoughts on the aim here or questions about anything cool well then let's get to the roadmap so the roadmap i just copied out the section headers how that takes us on a journey or on a narrative in our learning odyssey in a sense in the beginning there's an introduction and a theoretical framework for affordances is provided and that includes i know the font may be a little small perspectives importances and phenomenology landscapes and fields meanings and affordances and then we reach figure one which has a basic cognitive formula for automatic intentionality and figure two that expands on that formula slash comment format so that's a theoretical framework for affordances maybe could anyone chime in what what is an affordance in this uh context or in the cultural context or in the context of whichever field you're familiar with yeah so i guess just for a simple definition it's something that invites action or interaction and the the cartoon example is always a chair a ford sitting um but in the context of a social interaction it could be some environment that affords or that elicits our interaction maybe conversation or music making with another person great and it

makes sense to start the paper off with this framework for affordances because action opportunities are so fundamental to active inference and to action-oriented frameworks this is really what defines action-oriented frameworks is they focus on actions and the stimuli or the cues that lead to action so affordance is their opportunity the second section here is the neurodynamics of affordances so this is the integration of the cognitive and the neurosciences with the more behavioral ecological concept of an affordance computation representation and minimal neural models is addressed and also many of these ideas later papers include other or different depictions and this is already four years past so that equates to about 400 papers in between us and here yeah i should say um like this is probably the weakest section of the paper as it currently is formulated i mean yeah as you're pointing out it was published in 2016 and we started really working on this in 2015 and i mean since i did a whole phd focused on the free energy principle and uh what it what it entails and how it works and so on and i think uh i i mean i wouldn't disavow what we say here uh but i think that definitely our our understanding of the representational implications of the theory have been refined over the years but i mean this this is basically in a nutshell what we tried to pursue since 2016 so as you're saying like several papers have been published since um and the uh you know the the framework has moved a little bit absolutely richard yeah i think i mean definitely whenever we look back on our earlier papers we're always like i should have said all these different things but i i actually really appreciate this paper because you know for reading it just as a complete non-expert in a lot of these fields um the free energy principle is very very difficult to find an entry point into and sort of have something that i think is very conceptual and grounded in neuroscience i think a lot of social sciences could wrap their head around you know relatively easily and then to trace this as breadcrumbs to the more advanced um free energy principle stuff um i i really appreciate i thought it was a great entry point into this literature and kind of gave me a you know a foundation to kind of start to wrap my head around how these these concepts are being applied in that more mathematical context you're saying that it's one of the reasons why we chose this paper other than shannon suggesting it is for me at least when i can see the path in the literature well okay here's what they thought in 2016 which means in years leading up to 2016 and then yes it can be refined or represented or enriched in later work but seeing the breadcrumbs or the pheromone trail is how i get a calibration of the derivatives of how the ideas are changing through time and i found that um yeah one way maybe weaker but the most growth has happened here and this is one of the areas where the most growth can happen so this is why we continue to dip back into the

literature even longer than four years in the past i know it feels like forever though because sometimes there's a uniqueness to that cultural artifact that isn't just about how well it conveys what the relationship is between neuroscience and free energy but rather how well it conveys the what the researchers were working on in that intersection so we're thinking through other minds for some of us it was us in the past for some of us it was just another paper yeah and i will say this just really quick too a lot of the affordances that concept of affordance in sociology is very much grounded in kind of a pragmatic process philosophy you know whitehead the idea that you know reality reaches out to the other reality and so if you if if that is where our literature is grounded to be able to have something like this that's very conceptual to work through those philosophical ramifications and then begin to ask how would this fit with uh the free energy principle and i think vice versa as well having the free energy principle community kind of understanding some of these philosophical concepts and ideas to kind of ground more conceptually not so much mathematically what it is that their perspective is offering to other disciplines [Music] well said then we move to figure three now i found that figure three four and five make a really nice sequence and we'll go through them in sequence because it enriches one level of detail on top of one level of detail of a hierarchical prediction error minimization framework figure four diagrams it a little bit more clearly from a bayesian perspective and then five brings on another level of realism then we talk a little bit in this section about predictive processing and attention and that returns our attention to the idea of culture as regimes of attention from the title at this point if i had to rewrite this paper this would be two papers and this would be the split so i really think that the first part of the paper really uh tells a story about affordances uh affordances the affordances that are avail available in the human landscape uh you know in terms of their depend the dependence of certain kinds of affordance on convention whereas others are more just dependent on the kind of spontaneous you know action biology of human bodies and then the second half of the paper turns to this concept of attention and with with all this in place says okay so how is culture acquired in these what we're calling regimes of attention which are effectively shared patterns of allocating your attention to this rather than that as being salient so and this all is bit you can read this as kind of computational goffman uh for richard who's done comes from sociology uh yet it's an attempt to kind of spell out how what would a mechanistic model of you know regimes of like civic inattention and so on look like effectively uh when you're trying to bridge these different levels but you know i think there's a lot in this paper and this is a good kind of split point if you want to see what the paper is doing effectively nice very helpful for us

because it just reads as a turing tape to those who are reading it line by line but to know that there's a conceptual demarcation is helpful and regimes of attention i just think is a great term to capture it because it's kind of like well where's the regime it's everywhere and it's nowhere it just is the water that culture is existing within in this section we have figure 6 of a diagram of the looping effects or the rolling loops that mediate cultural affordance learning there's a discussion of skilled intentionality and affordance competition with some nice citations to other work about the skilled intentionality framework there's a section on shared expectations local ontologies and cultural affordances and we think a lot about ontologies here and about the words that we use to locally organize our thought and cultures this kind of multi-scale ontology that helps organize stimuli in a sense and the section closes with a discussion of shared expectation and implicit then the last section major section of the paper is the regimes of shared attention and shared intentionality reflecting gating abilities and and moving the loop regimes of shared attention and skilled intentionality before a conclusion so that is the roadmap that's how you get from discussion of the ecological psychological or behavioral inactive concept of an affordance then into the neuroscience of affordances and then as suggested by maxwell there's a bit of a intermission conceptually here before turning to a second half of the paper that has to do with the actual ways in which culture is scaffolded through these regimes of expectations and attention any questions on the roadmap all right here we are in the abstract which i often find helpful to read because i know that authors put a lot of care into the abstract because it represents to the in group to the field as well as to the out group to the reviewers to the editor to the reader it reflects what the paper is going to be about and people get surprised and even disappointed when the abstract is off base relative to what the paper actually delivers on they write in this paper we outline a framework for the study of mechanisms involved in the engagement of human agents with cultural affordances which we're going to define soon our aim is to better understand how culture and context interact with human biology to shape human behavior cognition and experience we attempt to integrate several related approaches in the study of the embodied cognitive and affective substrates of sociality and culture and the socio-cultural scaffolding of experience that's the first half the abstract then they write the integrative framework we propose is we propose bridges cognitive and social sciences to provide one an expanded concept of affordance that extends to sociocultural forms of life and two a multi-level account of the socioculturally scaffolded forms of affordance learning and the transmission of affordances in a patterned socio-cultural practices and regimes of shared attention the framework provides an account of

how cultural content and normative practices are built on a foundation of contentless basic mental processes definitely something we'll return to the acquire content through immersive participation of the agent in social practices that regulate joint attention and shared intentionality okay any thoughts on the abstract or questions before we turn to the box well box one i thought it would be helpful just to list out the terms of the different types of affordances then go through them what is an example of each of these kinds of affordances or what are the differences between these types of affordances so maybe we could start there if someone could raise their hand and pick one of these types of affordances natural cultural or conventional and then define uh what it means to them or an example of it yep maxwell um well essentially what we're suggesting is that well cultural affordances by the way um i i just noticed that my um my camera is frozen is that the case for everyone as well your screen looks dark you can either toggle the camera on and off or just reload the page um after talking i might try to rejoin just so the camera freezes um yeah so what we're essentially arguing is that in humans all affordances are at least minimally cultural in the sense that i mean i don't know about you but as i was growing up uh my mother told me not to slouch uh while i was at at the dinner table uh and so you know it's something as simple and natural as sitting um does actually come with a cultural scaffold and what we're saying is that there's basically a spectrum of you can think about the space of affordances with his with which humans interact as going on a kind of continuum from natural to conventional where natural affordances depend mainly just on the biophysics of the human body in its interaction with the environment whereas conventional affordances depend on these implicit or explicit sets of uh conventions or shared beliefs uh so you know an example of the latter would be um a king's throne right you don't sit on the king's throne unless you're a very specific agent uh so a king's throne doesn't afford sitting uh for everyone in the same way that say just a chair or a stool might afford there there are conventions shared conventions uh the kind of uh structure which affordances are available uh and which are not for a given agent cool richard yeah i was just curious in the neuroscience about like actually distinguishing that so if i if i see a cup my mirror neuron system fires and i begin to kind of enact the act of reaching out and grabbing it but then semantically i know that you know as a a lowly surf i'm not allowed to touch that is there actually any research showing the conflict between semantic knowledge about you know and self-related knowledge that i'm not allowed to do that with the automatic kind of reaching out mirror neuron system simulation of touching the cup that's really interesting i'm not sure i i don't know if there's any data about this particularly it would be very interesting though one



thought i had was some experiments i believe by michael von elk with hands grasping cups at incorrect ways so it's something when you're here and you're on c oh that's not how you're supposed to grab a cup well why not well because i can't rotate my wrist again to drink it so it's the wrong way to hold it because it's not really a cup of water if i can't drink from it in that functional affordance based sense but i agree it would be interesting and we'll follow up on that and stephen did you have any class yeah i actually had a workshop when i i was working with a group with um who with disability who um and i mentioned about affordances as part of it and partly having done this paper and one of the participants she was saying how when she sees a cup because she can't pick it up and she doesn't see it as a cup unless it's one that's got a handle that she can reach her hand into and do that so that kind of sense of things presenting is quite true and i like this breakdown because the i get a sense that the following on from the later work that maxwell did like natural affordances is like you know if there's a cliff it doesn't afford me to walk any further whereas cultural is a bit more of like this mixing sort of egotic kind of space where it's conventional it's like the the throne it's like it's not up for negotiation it's like you don't sit on the chair it's like a fixed rule there's no kind of inter-subjective piece going on so in some ways the cultural piece seems to have some sort of negotiation associated with it which is quite quite cool would it be accurate to say that that in a sense of if i i i had the simulation to reach out and grab the cup but then my semantic memory says no we're not allowed to do that could that almost be like a prediction error in the sense of like i'm predicting that i'm going to do this well no we don't do that precisely and the way that you would do that is probably by having a hierarchical model where basically you know so the the notion of hierarchy i mean it has political implications and we can kind of ditch those uh so i prefer thinking about it in terms of center and periphery right where like the the periphery is the the end of it that's closer to the sensory motor like interactions and the center is something more like you know what we would think of as the top of uh of a processing hierarchy where representations are progressively more compressed uh so if you look at the hierarchy from from periphery to center or from bottom to top what you have represented there uh if you'll allow me the language of representation for a second we can finesse it later but what's essentially represented there is as you progress towards the center slower and slower or more and more abstract regularities so you might think that yeah as you see the chair at a kind of more automatic uh kind of a sensory motor level you you see it as something that you could sit on but as that ascends the processing hierarchy at some point it registers an error as in like semantically i'm not allowed to sit on this because it's a king's chair so precisely

that's how that would work as and you can you can accommodate for this kind of conflict by allowing the system to have a layered or hierarchical structure and great point and this relates to how it's a spectrum of affordances so for example a violin when i look at a violin i have the biophysics to bend my fingers but i can't play violin so i see an affordance for something i could pick up potentially and not everybody may see that affordance however somebody else sees that and culturally they think oh i could play this uh song or i could play that song and so what song they see the device as being able to be used for is going to be part of their enculturation part of their education which builds upon the biophysics of the human body and that's why we have different dance styles in different places from different cultures even though everybody's elbows still move the same way sasha yeah so in trying to understand affordances um i tend to think of it as like the space of what is allowed but in walking through the different kinds of um affordances that we have um it becomes pretty clear that it's not just what is allowed it's very strongly gated by the top-down priors of what you think should be allowed in your space and it just kind of reminds me of this anecdotal understanding of how children interact with an environment versus adults how children would use or sit on objects that adults would never think of sitting on and kind of the the different ways that they see and use the world um because they don't have these top-down priors of um what one should and should not do and so that that's a nice way of um kind of circling back to the point of this paper that um everything is embedded in culture and you don't just start from kind of the biophysics of sitting and work your way up you're always going back and forth with this is uh this is such a great point thank you for thank you for uh pointing that out sasha like the the agenda here is to uh pursue this kind of excuse me this kind of cooperative turn in the anthropology literature which is to say that you know as joseph henrich who's one of my uh one of my favorite like anthropologists uh psychologists uh out there as he points as he puts it we've crossed the evolutionary rubicon in a way our our survival as humans depends on our ability to reliably access and use uh stored cultural information effectively and that's had an influence on the way that our biology has evolved so you know compared to chimps for example we have uh really weak digestive juices and that's because we've we've been able to offload the metabolic cost of you know having this extremely elaborate digestive system uh to our technology effectively you know so tool use the use of fire cooking technology has allowed us to process food in a way that then co-evolves with our bodies and takes the kind of processing load of the food uh off the body itself towards these cultural artifacts and practices that have co-evolved with our biology and that's really the the point is that uh all human

affordances are cultural affordances which exist on a spectrum from more natural to more conventional but human biology is a cultural biology and biologic and yeah culture is a biological phenomenon i think that the two are kind of irreconcilably joined in the case of humans and i'll build on that to describe the difference between the field and the landscape of affordance sorry i meant insurmountably you're uh irreducibly joy not not here uh yeah sorry oh yeah it works so the field of affordances are the affordances in the landscape with which the organism can dynamically cope and intelligently adapt so that's at an organismal level the field of affordances is like the field of play it's the field of action for that agent the landscape of affordances on the other hand like the landscape exists beyond me even when i can't interact with it directly and similarly the landscape of affordances is defined here as the total ensemble of available affordances for a population in a given environment and so the concept of chair just as sasha was bringing up the landscape of affordances of chairs includes things for little people for tall people for people with small hands for larger hands the landscape of affordances is at the population level which means it includes all these differences that are biophysical and developmental and cultural as well so the field is what one individual agent is able to interact with and that's sort of where the rubber hits the road of action-oriented frameworks and then when we pull back a level to this evolutionary perspective on action-oriented frameworks we get to this landscape of affordance's idea any thoughts on that yes stephen yeah i think that that way that things are embedded it comes back to that piece with the throne as well is where where that chair is for it to be ascribed as a throne that you don't sit is partly processed by what's going on around in that setting at that moment like if there's no one around and it's an empty room you may sit on the chair if you don't think anyone's going to see you but when the cultural kind of rules of the game are in play or if you see you know if it was a room where there's only children in there compared to a room where there was an adult watching so there's this kind of negotiation going um which is quite quite complex and i think previously without free energy i don't think there was a way to be able to take that many different parameters and sort of integrate them richard yeah i think what's really fascinating too when we talk about this is that um the role of language in the role of imitation and the role of just watching other people where i can walk into a room and know absolutely nothing about the norms values and beliefs that that constrain it but if somebody just tells me you know four or five words they can immediately bring me up to speed and kind of and to use the terminology that sasha was using earlier those top-down priors i can gain a lifetime of experience in 2500 words and watching four or five people seeing who's rewarded who's

punished and i think that i i i'm curious to see and we don't have to talk about this now but just you know rhetorically i'm curious to see how can you take in that notion of symbolic communication and quickly affect somebody's you know prior understanding of situation how do you fit that into a network of free energy or agents running off this free energy principle and how does that is it like a synchronization of oscillators on a network like how exactly do you model model that process yeah i'm just imagining you're walking with somebody to a meeting and right before they open the door to reduce your uncertainty about what's behind the door they say hey it's going to be tense in there or hey just so you know it's a non-smoking room and just like you're saying with four or five words the way that that person unpacks those words symbolically and the way that that communication channel is utilized to attune the agents on a narrative level that's what dictates how well the scene is going to go when the room is actually being inactive and so there's so much that we tell each other when we're leading into events that that does set the tune and that has to be unpacked within each person's own perspective and also one bonus word that's not here but umvelt does anyone have a sense of what that means but i figured i just throw it out there as they sort of complement to affordance well if i can just uh comment on uh richard's point uh before getting into unveiled i think the way that you can understand it from a free energy perspective is uh i mean andy clark i think and andrus robstorff has have called this top top effects basically uh passing information from the top or the center of one's hierarchy to the top of another directly through these kinds of uh of linguistic tokens that we're exchanging i think that's uh that's a powerful intuition what we've done in more recent work is extend that to a kind of horizontal top kind of communication where it's literally destruct the shared physical structure of the niche whether it's in the in terms of shared artifacts like stories that we uh create and and share with one another that so these uh effectively these uh parts of models that are offloaded onto the physical structure of the environment can also come and have these uh kind of uh top-down structuring effects where we're we're effectively installing shared environmental priors at into the top levels of our hierarchy through enculturation nice and how about maxwell what's your take on him belt i mean i think you know so i think those of us uh you know working in uh more specifically on ecological psychology we'll have some beef with the way that i kind of uh just generally put all of this together but i think like you know affordances umvelts uh i'm sure many people are not going to agree with me so you know feel free to discuss this but that they're they're all perspectives on the same kind of thing which is that if we're trying to talk about you know the meaning making of organisms living organisms that share a world we

can't uh we can't just not talk about the first person perspective and the way that meaning appears from a specific vantage point and that's sort of what the affordances and umvelt stuff gets at uh and effectively what we tried to do in in this paper and that's why i think it's it's the kind of contribution to the broader project of neurophenomenology is we we start from the kind of first-person perspective so we we're kind of trying to cash out you know some some subtle social effects like enculturation in terms of learning affordances and then we ask the question what's going on under the hood and as it were around the hood uh for that kind of affordance story to be cashed out in mechanistic terms uh so umvelt is a term uh by a biologist i think it's pronounced von ushakul yeah and he uh i mean it's it's essentially it's very close to the uh the uh affordances story although it's it's more directly centered on um the perspective of the organism i mean so gibbonians i should say like uh you know the ecological psychology was introduced by uh you know jj gibbon the famous psychologist and a lot of ecological psychologists have uh take issue with our use of affordances here uh i think for interesting reasons the so the for for ecological psychologists the idea is that perception is the direct perception of a possibility for for well it's it's the directly we're directly picking up information about our environment like so they're trying to emphasize that we don't have to go through this elaborate kind of inference process there's a sense in which like we just directly read off the regularities in our sensory array uh what what what is possible and what is doable and it so in gibbon's original formulation this is what affordances meant what we're doing is more drawing on tony camaro's reinterpretation of this concept as affordances 2.0 he calls it where an affordance is a relation between some relevant features of the environment and skills ultimately you know expectations and skills had by an organism so this is already a a more inactive embodied reading of affordances and i mean we've gotten into some very interesting disagreements with people from ecological psychology precisely about this difference in interpretation so i say this because umvelt is very much focused on this kind of affordances 2.0 perspective where what we're trying to do is really capture what the world is like for the organism that inhabits it from its perspective this is also where i think it connects with phenomenology which is also this kind of more systematic attempt to describe things from the first person perspective that okay alex first and then shannon yeah thanks um as usual i'm trying to connect uh to my domain is the engineering domain it's more like a question about could we consider importances from some kind of functionalities view as for example if i need to drive in a nail the hammer the provide provides me affordance to use it but if i don't have a hammer i could use even microscope to do the same only one or two times but anyway for my

engineering task it could be enough and if i see a chair and it's provide the affordance to sit it's like a basic function of the chair to to to sit on it so from this perspective how it could be considered like when agent need to do something and need some kind of functionality and this functionality or maybe service to do it how it's possible link oh absolutely this has been pursued by a lot of work there's a really cool group based in um in the netherlands uh eric rietveld and julia keverstein in particular uh work from this perspective where eric uh eric ridwell's work is in architecture and affordances so he he really tries to bring this whole affordance's perspective uh into the design of the spaces that we live in uh so what you're saying is totally uh consonant with uh at least some really interesting uses of the concept of affordances in the literature since the early 2000s cool shannon i think yeah so i'm just thinking of this this field landscape difference and umvelt it's really helpful considering the perspective of the agent you're investigating especially um so i'm in music neuroscience so we can learn a lot about people or brains interact with music by putting a person in an fmri scanner and playing some music but that's not exactly how we interact with music in our day-to-day life maybe now more so because we can turn on our ipod and just sorry iphone we don't have ipods anymore um and listen to music alone as this um solo endeavor but in general for the history of human evolution music has been this really interactive participatory activity and in order to understand um from where i am how we interact with understand or process music it really is also a question of how we understand and interact with um other people and it's so when we bring in these affordances we can have an affordance for music um in a certain field that's you know us listening to music alone but as soon as the landscape is is a little bit bigger and maybe our our small field incorporates one other person or one more other person um then the perspective that we need to take needs to change to adapt the perspective of each of those individuals or of that musical sort of environment as a whole that's such a great point i don't know if you're familiar with this paper that i just linked in the chat it's by anna anderson and tom zimkia uh called exploring the multi-layered affordances of confor of composing and performing interactive music with responsive technologies and it was one of the first papers to pick up our cultural affordances work and in precisely the context that you're proposing shannon like um what they're saying is uh our concepts need to be enriched with a kind of intermediate scale concept so their idea is like when you're when you're performing live electronic music there's something intermediary between the field of affordances that each individual is coping with at any moment and this evolutionary level kind of landscape of affordances or this population level uh landscape of affordances where there's you know the

there are special affordances that open up in this uh performance uh context kind of a a meso scale in between the landscape and the field uh i forget the precise term that they that they introduced but uh yeah the music is definitely one of the cooler domains of application of this uh this stuff nice sorry all right shannon and then just saying this is really great thank you cool sasha um yeah just a brief comment to follow up on what alex said um uh from the engineering perspective you know if you have a hammer everything's a nail it kind of made me think uh about crime where um if you need something and you're not um inhibited by uh law or social norms you can just take it maybe that's a uh overly simplistic understanding of why people commit crime but in in that context if you're not inhibited by laws then your affordances are quite different than if you are okay and perfect richard you're uh yeah in terms of the the field versus the landscape would it be accurate to kind of conceptualize this the field of force is kind of the episodic knowledge of the individual so what are their their personal learned experiences their personal body capabilities kind of this really episodic self you know um form of memory where landscape of performance is more about the semantic knowledge the if there was a way for me to kind of look at the population and ask what are all the skills and habits that this population has that would be the landscape if i was to drill down on one person how well do you know how to ride a bike that would be more of your fuel performance precisely that's precisely the idea and the idea also is that not all affordances are equally salient at a given time but they're still there in a sense right uh so i don't know like um 24 7 uh coffee shop there's times a day where it's a salient affordance to some people well i was gonna precisely give a a food and coffee related example uh yeah i i drink coffee in the morning but i tend to uh have insomnia if i if i drink it too late in the evening so there are times in the day when coffee affords drinking more than other times in the day and so you know it it's still the case that at any moment i could go into the kitchen and make myself a coffee right so the field of affordances has more to do with those those affordances that are soliciting me at a given moment uh you know it corresponds to the more the phenomenology of interacting with affordances whereas the landscape is more an anthropological or sociological statement as as richard you were just saying about the kinds of things that a population can do in a given space it that you talk about that idea of sailings in the moment because one thing we talk about a lot in sociology is is identity and so if you activate my identity prime me to think of as a professor i'm going to notice certain things in environment differently than if you prime me to think of my identity as a husband or a son or a brother um so that kind of then again but identity means you know my identity as a professor may be different from any one of your identities as a

professor based on socialization and culture and disciplines even as well right so okay interesting thank you yep and just one point to tie it back they write as high digger famously argued it is only when my smooth coping breaks down say when i run out of coffee or when the cut breaks that the objective properties the cup becomes salient present in perceptual experience at all and depending on if you're at home you run out of coffee maybe you go back to the coffee machine you set up a new brew if you're in a restaurant culturally maybe you raise your hand or maybe you whistle or maybe there's some other thing that's done culturally to signal that somebody else should see that as their affordance to come over and ask you what you need so the landscape of affordances is a very role specific uh it's realized through specific roles but it represents that population level and that's so yeah lots to say here that was an excellent discussion let's turn to these first two figures that are cartoons about basic cognitive formulas and a little bit more of a full formula so in this first three panel cartoon i'll read out the captions and then maybe somebody can give an example or a thought the first caption is the person thinking i think they think i think the second panel says what would others expect me to do here and the third panel says i inferred that they intend that i should think feel or do x so what is a thought or an interpretation or a punch line to this cartoon what would mommy want me to do in this context and i you know it's i think this is sociality is kind of scaling that up to an anonymous kind of third or other or that is you know yeah this kind of anonymous sociality is scaling that like kind of you know parental expectation thing up a bit yeah it's like peer pressure made me do it well yeah they move your joints for you well no but people expected me to do it well just because they expected you to do it even if that's true why did you do it and it comes back to this that is an account of what peer pressure is from from this perspective even if the actual biophysics of the joints are not being pushed by other people and the the point that we're making here i mean it's really developed in our bbs paper uh thinking through other minds but that it's that human thinking really is not just thinking about other humans but thinking through their own perspective uh and that much of our social interactions actually are structured by the beliefs that we have about other people's beliefs about our own behavior and beliefs so you get this kind of nicely recursive intentionality structure so that that's like the simple kind of you know the basic cognitive formula that we're calling it but the i we think that this really applies to almost all human thinking we're really always kind of trying to understand the situation uh from the perspective of you know shared beliefs about what's appropriate in that situation which which implies the ability to kind of perspective take and to think through the minds of others yep when someone's talking to hear themselves talk



as we say it's off the rails because it's not in feedback with other minds but when somebody is attuned to the narrative and the culture of the local world the local actual interaction not just the hypothetical then there's the opportunity for communication to be targeted and direct and actually move towards shared goals and under shared values as well instead of just individuals biophysically running their mouth which is an affordance that we also have so how do we move our communication patterns towards this higher level um instead of just using the opportunities that we know that we can also have any other thoughts on figure one figure two is related is that we talk a lot about perspectivism and you know we go up to higher and higher levels but i think you see a point at which you can only take a perspective on what you think to a certain level it gets too complex too multi-sensory too many inputs and it is that kind of phenomenological embodied feel for what you need to do which is not something you even have a perspective on that idea of the flow there's this info inferential kind of approach as well as a perspective approach which just comes from just feeling what you should do and i think this this kind of shows that because i think you can only go up so far and take perspectives on things when how you act in the moment sometimes you just have to be there and feel it yeah and we can't perfectly emulate what somebody else wants and it's just not that simple we're in the bodies that we have with the affordances that we see and so to try to over engineer what the other person wants or should want or should want to want it's uh it's a good thought experiment and it helps in a lot of situations but it is not the end game of relating well it might also explain some of the difficulties that we're having currently in our political landscape uh so i mean as a as a canadian i'm very sold on the idea of multiculturalism and diversity and pluralism but i think these are not intuitive modes of sociality precisely because uh they oblige us to multiply the different perspectives through which we're trying to you know relate to a shared social world um and you know like uh things like dunbar's number i'm sure you're all familiar with dunbar's number right if you if you uh basically plot the size of primate brains against the size of their social groups there's almost a linear relation and if you extrapolate that to humans we would you know live in groups of about 100 150ish and you know the the that's that probably has a lot to do with you know the the kind of sweet spot where it's easy to kind of share uh you know a bunch of uh beliefs and norms and patterns of behavior with the group and things get more and more complicated as the group expands and diversity is included which is not to say that diversity is a bad thing by all means i think uh you know we should move in that direction but i think it speaks to some difficulties that we're seeing politically that it's not an intuitive mode of sociality we'll return to that when we can and here in figure 2

we're almost rephrasing figure 1 a little bit more from an improvisational perspective so the left panel the person is thinking how much can i improvise here the middle panel is what do i know they know about me right now and how much room to improvise does that give me the third panel reads how much can i improvise here given what the local cues tell me about what others expect so a lot to say and think there's probably other ways that these recursive questions could be formulated but is there anything that we haven't addressed in one that people might want to address here cool this just reminded me of fellow jitzer yes yeah um i was i was thinking that um maybe in the adolescent period um we are kind of rebel we are rebels against society and trying to find our identity maybe in other words um minimizing our self prediction errors so we are like trying to test these norms um by being uh level so actually by impairing uh what others intend that i should think and feel maybe if i do the contrary learn something about me how can we deal with this um like landscape affordances and cultural affordances or even what our mom is saying and what we are really trying to find out about ourselves nice maxwell uh oh well uh yeah uh uh so i thought that was very insightful uh thank you very much uh for that very uh very interesting um well i would have just added that so the point here is to kind of say well it's it's not it's not merely just about you know inferring uh what others think i should do it's it's really more about the situational coping right so it's about like learning different contexts and learning about what's allowable in all of these different contexts and then being able to juggle that and to reactivate the right constellation of you know semantic knowledge for example or rules and norms that pertain to each and and also then you know moving from one context to the next and seeing what kind of flexibility that allows for ultimately uh because you know human reality is improvisational like it it's it's about kind of you know uh coping with with a situation given the constraints of that situation in real time um yeah yeah it's a controlled novelty question like a lot of improvisation it's a on the edge the leading edge of the trade-off between explore and exploit where successful systems are at and so the teenage rebellion if the teenagers just mindlessly fell into the line of whatever was happening before maybe in some niche that was unchanging that would be a successful evolutionary strategy but the niches are always changing not for the least of which is ecology but also improvisation has this always moving dynamic and so the teenage rebelliousness is there's the element of the individual coming to precision about who am i and who am i in this broader sense and then at the higher level there's this explore exploit where some element of contrarianism it comes across like a negative word but if you've ever had contrarian models then you couldn't know if you were just in a very very local optima or

whether there are other realms to explore so there's there's so much there and the details are all in actually formulating these specific situations and understanding what are the affordances what is precision uh doing here so maybe in the next bit we can pretty briefly look through figures three four and five so maxwell it will be helpful to have your perspective on these but let's start with three so figure three is a hierarchical and maxwell maybe could you give a shot on what was being summarized here sure um i mean we're just describing uh bayesian predictive coding here most uh predictive coding schemes are hierarchical as we were just discussing uh above where um basically what i mean this is more explained in i think figure four or five but essentially what this figure in particular is supposed to illustrate is uh the kind of bi-directional message passing that's implicit uh in in all of this so as we were discussing you know the top or the center uh is where the more kind of abstract or slow regularities uh are represented and they they effectively the the top layers provide context for the bottom layers in the same way that slow regularities provide the context or the embedding against which uh you know faster regularities unfold well these slower regularities or priors provide context for the the ones below so what you have descending or you know coming from the center to the periphery is effectively predictions about the data that each subsequent lower layer should be receiving and the way these schemes work uh is uh i i think consonant with the uh you know that so essentially if the brain doesn't have to process something then it doesn't uh the brain is a lazy organ so the the thing i i often say to drive this home is what was the color of the last door handle you opened there's a large there's a high probability that you didn't register that at all uh well because there's no reason to right like you just smoothly coped with with the door what you needed to do is open the door to get to the other side of it and you know there was no reason to even register the color of the door handle so what with this kind of framework says is the signal that the brain is processing at any time is the discrepancy between the data that it was expecting to register and the data that it actually does and this difference is called a prediction error and uh if you can go just back to figure 3 for one second um i just want to make the point that well so uh this relates to base in the sense that descending messages carry prior probabilities so essentially you can think of this as the base rate for a given phenomenon independent of the data that i'm collecting and then the prediction error kind of combines that knowledge with data so it's something like the likelihood if the prior is just the probability of an event that i'm considering the likelihood is the probability of that event given some data that i've actually registered effectively this is why these frameworks are often called bayesian the ascending messages carry prediction errors which

are likelihoods and the descending messages carry prior probabilities which which are the predictions that the predictive brain and so on are all about perfect and then from here uh we move on um a specific hypothesis the predictive coding i'm not sure 100 the degree to which we're still committed to this functional distinction but the idea is that um in at least in the brain they're going to be two kinds of neural populations uh the ones which encode effectively yeah these generative model units which are the ones that are encoding the the kind of base rates about the phenomena that you're interested in and then another uh functionally distinct sub-population of neurons that's responsible for computing the difference between the data you expected given these predictions and the data that you're actually and again the unexplained signal that the part of the signal that wasn't predicted by uh the descending uh messages is passed upstream uh in the hopes that it'll be explained away at some point and also one note that can help understand this is this caption in the empirical base framework the system can then use the posterior obtained from one iteration as the prior in the next iteration and so the empirical base framework is in a way what it cuts the gordian knot it prevents us from just saying well it's just simply priors all the way up and we start somewhere by starting our prior for one level with just empirically what we're getting from the posterior at the lower level at that time point as a starting point and so this can be instantiated in physical not infinite systems because there's a possibility that empirical phase just as it's used in data analysis in a variety of fields you can use the data to inform a prior and then have a generative model that starts working from there yeah precisely and what bayes does in this context is just provide a way to optimally combine what you knew before sampling any data uh with what you've learned from that data uh so as you're saying uh what you're essentially doing is multiplying your posterior so you calculate your posterior by multiplying your prior by your likelihood and then uh normalizing uh which can get a bit hairy at times that's effectively what you're doing and let's just look at this example so here we have the same architecture as figure four so we're building it out figure three this is a uh it's a graph it's just nodes and edges it's the pure representation now we're thinking about a little bit more of a plausible neural architecture and maybe in some brain regions it is this way and maybe in other brain regions it isn't this way but this is a plausible architecture and now we have an example so what is being shown with the dog well this is just to re-illustrate the center versus periphery distinction uh well so the the image of the dog has just undergone a fourier decomposition which basically means that you're kind of separating out the different spatial frequencies uh that are used to generate the image and i mean the hypothesis is always sort of the same is that the closer you are to the

sensory periphery the the more you're dealing with faster changing regularities uh so high spatial frequency information is represented at the bottom or the periphery of the hierarchy and as you ascend the hierarchy what you see is essentially lower and lower uh spatial uh frequency information uh so this is the kind of idea here is that like the the the fast regularities are represented closer to the sensory periphery slow ones closer to uh the center i hope i said that right and um and and yeah uh most sensory modalities are gonna be uh deconstructed in this way i mean you know for the visual uh brain regions uh you start off you know detecting uh essentially lines and you know like grating patterns and the more primary us visual areas like v1 and everything and then as you ascend the cortical hierarchy what the neurons become sensitive to are groupings of these patterns so things that change at a slightly slower time scale i mean if you did one of these fourier decompositions to an image of the face what you'll see is that the gross morphological features of the face like you know where your cheekbones are and where your uh you know jawline is are much coarser than the fast changing things in your face like your eyes and your mouth uh and so again you you might assume that this is represented hierarchically uh in the brain um and let me see i'll add one more level to the specific visual example because it's in the specifics that we can see what we're getting out of this framework so low spatial frequency is like a blur filter it's a very uh it's averaged out over large regions of visual space on in contrast the very high spatial frequency it looks grainy and that's kind of like turning the contrast very high up to over emphasize differences between neighboring pixels but when you do that you lose the global structure so we know that from playing around in photoshop well it would be really cool if we could just see the world in total high resolution and just optimize this with one level but it turns out that in the retina the cells are activated even before the lines detection it's like light first dark it can only detect whether it's on or off extremely locally and so how does that signal which is ultimately an extremely local contrast signal how does that percolate up in the context of deep priors about regularities in nature to look at deep mind look at the images that are produced in deep dream these regularities that appear natural they're not just any kind of natural they're human visual niche natural like grass and clouds these kinds of things have structures that certain other kinds of repeating uh like a grid just doesn't appear in nature and so this is a structure by which the hyper local contrasts to contrasts literally between pixels in this image can be passed up and you can end up getting this full spectrum for your uh resolution on the image and then to add one more level of detail what we experience isn't any single one of these levels because we experience a world in full color though there's no uh color receptors at our periphery and also

a world that's at full resolution though we have very very different resolution as far as the pixels are uh the retinal detection units are concerned in this phobia in the center of the eye versus the periphery so that to me is demonstrative that we're experiencing our degenerative model of the world which is that the world has similar resolution no matter which way my head is looking and there's color everywhere and then subconsciously or sub personally our eyes are engaged in an oculomotor informational foraging task and where do the eyes saccade it turns out that they look where the most informative place to look would be given the structure of this bayesian algorithm so so many things that are suggested with just this architecture and there's so many aspects of the signal processing cascade and especially the top down priors that aren't reflected by this idea that the eye is a camera and the brain is a cpu and it's just doing image classification like a convolutional neural network there are convolutional elements to be sure in vision but they're playing out in this type of framework not in a tic-tac-toe framework yeah absolutely couldn't agree more with that that's uh that's a great point i mean uh the i think an often overlooked element of this picture is that it's not just saying like this is a criticism of bottom-up passive you know a passive picture of the brain that just kind of bottom up aggregates uh you know simple properties until you get something like a percept it's so we're not just saying that this is a top-down approach it's the bi-directional processing cascade that really gets you uh where you want to go ultimately and you know as you were just saying uh dan like you know the the experience that we're having implicates all these different levels and it's precisely because this whole multi-level architecture is implicated at all times that you get this kind of rich uh you know equally multi-level experience cool maybe we can turn to this last figure for the last section of our discussion so now we've built out this bayesian framework from the previous images first from its most austere framing to something that looks a little bit more naturalistic and now we're going to bring it back to our discussion about culture and about expectations norms and affordances so here we have the same structure where we have predictions moving down and prediction errors moving up or in or out these are just spatial metaphors and there's only in and out there isn't top or bottom and then we're adding a few more levels of details to the model the first is this blue and that's the precision weighting and so that's entering this hierarchy at a few different places and it's tuning the temperature so to speak on the precision so how much should we value how much stock should we take about this uh information that's coming up although there's a lot more nuance there and then also there's a top-down modulation of attention and that's where this regime of attention enters so the precision story is really key to all this and you know honestly

as i was learning this framework uh you know like five years ago initially i was hoping that the precision stuff wouldn't be so central to the argument because it's really complicated but uh it turns out that it you get you get some of the most interesting effects from uh the precision stuff so it's it's it's a little bit it's a little bit difficult sometimes to wrap your head around but i think the effort is worth it the the idea is just to say at every level of the hierarchy the the brain doesn't just register a signal but it evaluates how confident or reliable it is in that signal or how reliable the signal is and this is what this precision construct effectively uh quantifies and you know uh dan you were relating this to temperature informally that's correct uh a more intuitive way perhaps of framing it is as volume so it's sort of like the the brain's volume on each of its different kind of sensory streams so sensory and predictive streams effectively and yeah so uh precision waiting is what uh allows you to according to this story at least arbitrate uh different sources of information so for example 2018 2019 i spent a lot of time in london and london was surprisingly foggy um so it it so far i mean as a canadian i never experienced anything like this i think once when i was a child there was just so much fog and there were days when i would come out of my flat in london and i couldn't see two feet ahead of me there was so much fog and so uh you know in in circumstances like that what you learn to do is to rely on your other senses like you're hearing for example so there's a there's a story about the way that the brain is kind of dynamically it's sort of like a dynamic balancing act in a sense the brain is trying to see which which of the signals that it's dealing with are the most reliable um so i'll stop now richard has yeah yeah so and this is the the criticism i get a lot uh when i present at conferences and even when i publish on ideas related to this in the sense of you know if i'm let's imagine i'm a master burglar and i i walk into a neighborhood and i'm looking i'm chasing the neighborhood to find out what i want to what i want to break into which house i want to break into and so i have these priors i had the sensory information highly motivated to steal things and then there's always that old guy in the back who raises his hand and says that's really cool so how do i test any of that how what do i measure what variables do i need to collect and if at the end of the day all you're giving me is a more refined version of what i'm already talking about who cares and i i my gut tells me that's wrong that i mean as we clean in all these different disciplines we can go back and forth and translate you know our concepts back and forth and see that we're all talking about the same thing but if it doesn't lead to something empirical that then we can then start doing critical tests and show this version of the theory is wrong because my predictive priors or whatever would have looked differently if it had been that way so i i guess like to me that's the thing that i i'm always struggling with it and

i'm hoping i can get some insight from everyone here is how do we translate this figure six into an empirical test that i can write up for an nsf grant and then gather data and show my theory is right parsons from 1950 was wrong right or something that's a great question um i think the where this has been done the most extensively is in psychiatry and this is why you know uh this is what really drove the message home for me that the precision story was so central to this is that you can understand a lot of uh psychiatric conditions uh in terms of uh you know altered precision weighting profiles like uh i think a kind of well understood example has to do with the cycle the course of schizophrenia uh so we think that and phil corlett uh has great work on this uh i mean several people uh have worked on this uh from different kind of perspectives but all kind of relying on this construct of precision and backing it that then backing it up with fmri data so the course of schizophrenia for example it's thought to maybe start out as a you know a break a physiological break in your dopamine signaling so say d2 receptors start to misfire for no reason so they're generating uh what we would consider to be a spurious prediction error so like a prediction error that isn't really tracking anything and initially this gets resolved perceptually in the lower levels of the hierarchy it's kind of contained through hallucination like the brain is generating a prediction error signal and what it what it ends up doing is underweighting uh its priors are overweighting its prediction error uh because that's essentially what's going on like there's a breakdown and these prediction errors that aren't tracking anything are given too much uh importance and as the condition worsens they're given more and more importance because they're consistent like the brain is just consistently generating this prediction error and at some point it gets so intense in the brain that it starts to be resolved through learning rather than through a hallucination so then you know for those of you who do a little bit of research in psychiatry you'll know that there is a typical course of schizophrenia that starts off as a more kind of hallucination experience where people are still to some degree cognizant of the fact that they're they're undergoing an abnormal experience and they're hallucinating but late stage you know uh schizophrenia has more to do with uh persistent delusional beliefs and now your actual cognitive structure is adapting to the error signals that you're generating and learning from learning the error essentially uh which is not tracking anything in reality because it's just due to a breakdown from that point of view uh to answer the well to to address the points that you just raised richard the i think the uh like a lot of the power of the approach in terms of proof of principle comes from psychiatry these predictions uh and precisions are are computed by parts of the brain that you can identify and so phil corlett i think in particular has done uh some really



really cool work uh on um on hallucinate uh on the uh on schizophrenia and the uh effectively the kind of uh presentation of schizophrenia and the chorus of schizophrenia very cool work from that point of view it feels at yale uh for those of you who are interested cool um steven and then we'll return to the unique predictions question yeah i suppose just adding to that um this this challenge of things happening over time like a time series of things happening rather than kind of some snapshot in time which i think traditionally a lot of research sort of does that even what maxwell was saying there is what's presented as a pattern over time if you took one moment it might not show that and i think that's where it's kind of hard to analyze that that's why these models that they create show how things pan out over time on a low dimensionality and then you can get something from that and i'm kind of that does present a problem about real life scenarios and i'm my i'm interested in whether maybe approaching it from the direction of affordances and the environment and working backwards to how people are tuned to an environment might be the way to look at the more macro because it becomes so high dimensional if you take someone at the individual level but i think this is up for grabs let's bring that back to the question about crime uh by richard so let's think about the shared expectations so the shared expectations might be people don't walk uh onto someone else's private property so in that case even somebody who's highly motivated and has the tools the affordance to do a burglary there's still uncertainty about what's inside the house now let's imagine changing the norm so that people are have wide windows and you can see exactly which house has which uh equipment now somebody who's motivated and with the affordance is enabled by a cultural norm that allows passing close to the window as well as an architectural affordance to see inside the house they're reducing their uncertainty about what's inside of the house and that is going to change not just that agent's behavior in their own field of affordances but also some macro trends and so this is in one sense this is looking at within the brain these are different neuronal populations and that would be precision could be played by uh dopamine and by dopaminergic synapses like maxwell's talking about and personally i find it fascinating that so many stories about reward like oh well gambling is about people trying to maximize reward or bank robberies about people trying to maximize reward really is it though or is it about people optimizing their precision or perhaps even operating under maladaptive uh precision regimes so it doesn't have just because it's an optimization and a precision oriented framework doesn't mean that aberrant events don't happen so within the brain level we can think about that type of precision but then also socially what is the feedback that somebody receives along the way that have has other people give them feedback to keep

them uh on the straight and narrow as it were and then how do we as cultures define what that path is are there on-ramps from the fringes of society that help onboard people into a healthy and cooperative way of living or do we push people who even deviate in some way out of the center where they're even more likely to deviate further from the increasingly narrow perspective of the center so there's really so many levels that this can come into play at so i think just because of the time let's have a few closing thoughts from anybody who would like to share and then also please through text if you're listening as well as participating submit us questions or topics or quotes because we have a bunch of quotes that we've selected and some other things that we felt like we could have spent more time talking about but we'll save that for next week with this first week's podcast being the figures the definitions the on-ramps and the accessibility from a lot of different fields and then the next week we'll bridge out even further so any closing thoughts from our shannon i think that's a really good point that question of how do you study this you know how how is how do bring this entire story of the free energy principle or predictive processing or act of inference and actually study it in a social context like in a group of people so not just the stationary burglar who's alone in a neighborhood at night but in the context of a group of people who are interacting and i think that would be a really great topic for part of next week or like when we revisit and just you know a plug from music that's a great uh little model system to study these kind of interactions um and starting from western music just because i grew up here and that's kind of music i've interacted with but if you have an ensemble you know who are the leaders of the ensemble who are the followers and you can kind of manipulate that um really study how you can play with changing the regimes of attention or changing who in the group dictates uh what the particular field of affordance is that this piece of music needs to play um and there are a lot of really exciting and fun ways to get creative in the lab to study these kind of social exchanges and maybe put some of these um super complex mathematical models to like bring it give it some empirical testing great richard anyone else can raise their hand as well yeah uh building off that i think that's a great idea to talk about people in groups and you know the things that i've seen coming out with with hyperscanning and the ability to have two three four or five people all being scanned at the same time whether socially interacting and and then asking okay so what is this mathematical model of the free energy principle or active inference or whatever embedded an individual how does it what i mean does it come down to a couple of oscillators i mean that's the thing i keep coming back to you is this right this is just another version of coupled oscillators and if that's true then the notion of how do i empirically deal

with this is because these coupled models can deal with 10 15 20 1 000 people interacting that our current concepts in sociology we can talk about abstractly but no one can really figure out a way to measure it so i think that'd be really interesting to look at the intersection between the models our our current scanning abilities and empirical design and the theory and really see where they what are they talking about and what is a more unified operationalization of these concepts in light of this technology and these models what would that look like in terms of experimental design that's really exciting um i'll i'll try to rejoin uh because my camera crashed again after this comment but i wanted to say to both of you shannon and richard if these are lines of research that you're interested in exploring this is definitely something that my group is very keen on exploring uh i mean you're talking about hyper scanning i really think this is the direction for uh we're working with uh a few people now uh so robin murphy uh santiago castillo uh in particular are all people who do eeg and fmri hyper scanning and uh the the new kind of thing that we're trying to do is uh to combine multi-agent modeling with uh multi-brain neuro imaging you know because the problem with that i have with social neuroscience is that we're and i'm sure those of you who do social sciences will concur they're what we're essentially doing is looking at the brain with this really deep level of you know inquiry we're we're looking at like hundreds of different uh sensors on the scalp or whatever and so we have all of these this brain data that we're looking at but really when we're looking at like the kind of social measures that we're using it's it's these kind of toy measures like empathy or whatever that are supposed to tell us like something broad about culture and the problem with that is that we're effectively flattening all of the causal structure and complexity of the social world onto these kind of toy dummy variables that are supposed to tell us something really deep about uh you know social interactions and uh by combining this multi-scale perspective opened up by the free energy principle with uh hyper scanning and other uh social neuroscience techniques we're hoping to really be able to uh talk about and understand and model the social world in all its complexity while also looking at uh brain dynamics so i have people for example interact in ecologically realistic situations for humans uh and then see stuff about you know like uh so edwileck for example if you don't know her you should absolutely check out her work um e-d-d-a-b-i-l-e-k edward she's currently carl fristan's postdoc uh and i think she's also at mannheim in germany and what she's developed is uh effectively a measure of brain to brain synchrony or uh brain tumor real-time brain-to-brain information transfer and you can get to this by uh looking at basically real-time interactions between uh you know engaging in some kind of shared participatory task like you know

aligning gaze direction or whatever and simultaneously recording their brains as they interact and i really think that this is like the golden way forward so if any of you actually are interested in pursuing this please uh send me a dm or something and we can get started perfect and i'll just close i think with one short note on flattening versus deflationary theory because it really does come back to what are we doing here is this another metaphor and it's just pendulums on a table um or is there something actually more here and flattening maxwell was suggesting is when we're just projecting that's psychological as well as a dimensional term we project onto simple constructs we want to have a simple concept of empathy and in a sense by projecting into that representation at a lower dimension we've created a world that isn't necessarily reflective of the whole system in uh in contrast with a flattening account we can have a deflation a deflationary account where what's happening is really communicating brains it's really just those brains communicating and we're deflating away the extra the second level where is the empathy where's the regime where's the attention the answer is it's the whole system and it's not just the whole system meaning the person it's the whole system with the niche and the culture and so it's a lot to grasp grasp with because it is a multi-level system and we can design an experiment that isolates a certain component but we never truly have isolated from the evolutionary context or the developmental context and so the goal to pursue inflationary rich accounts that provide all of these humanistic things that we want to see like inactive embodied in cultured cognition rather than flattening it onto something and then saying well maybe they're the same between different cultures we can actually go into that richness and look at what we actually have and stephen with a closing thought yeah thanks i totally agree and i like this idea of being able to feel where we look this idea of feeling texture it would be sort of analogous to that because you have empathy and someone like i work a lot in the development field and community work and you get this idea of empathy but there's no time variation and empathy it's just like dump there resistance and it's like well how is that changing moments moment isn't there's none of that no one really thinks about that how things change over time so i think this this is actually really useful it's clarified a lot of stuff in my head actually just hearing that discussion i think it's a different way of thinking looking normally all that granularity would be seen as a problem but actually it can be seen as data which is really beautiful maxwell first author last thought i just wanna it's less of a thought than a feeling i just wanted to re-express my gratitude to everyone in the conversation um this is really exciting awesome and uh like yeah i i if anyone does want to develop these ideas like i was saying please get in touch with me and uh our our group is very keen on collaborating

with the we've established a a multi-site international network of research called the nested minds network uh we're gonna give uh we're gonna launch the website pretty soon but uh for those of you who uh are looking for international uh collaboration and so on please get in touch with me uh like what we're trying to do is build a community uh around this and you know this podcast is definitely part of the consolidation process and uh yeah so thanks thank you all uh you're all really cool and this is really fun thank you maxwell we built the on-ramps together that are bigger than we can build individually and build the on-ramps we want to see we build the culture that we want to see and that's a lot of work to be done but it's empowering to know that together we'll make it happen okay thank you all for the live stream thanks for listening and i will end the live stream